Hao Zhu

PhD candidate

Disciplined Optimization Group, Neurorobotics Laboratory
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— Education

2024– **PhD in Computer Science**
University of Freiburg, Germany

Thesis: *Disciplined Optimization in Computational Science*

Advisors: Prof. Joschka Boedecker, Prof. Moritz Diehl

2021–2023 **Master of Science in Computational Neuroscience**
University of Freiburg, Germany

GPA: 3.8 | German grade: 1.3

Thesis: *Deciphering Decision Making with Inverse Reinforcement Learning*

Advisors: Prof. Ilka Diester, Prof. Joschka Boedecker

2016–2020 **Bachelor of Science in Computational Chemical Biology**
Nankai University, China

GPA: 3.6 | Average grade: 89.4/100

Thesis: *Identification of Functional Residues in the Human Protoporphyrinogen Oxidase with the Network Model and Site-Directed Mutagenesis*

Advisors: Prof. Xin Wen, Prof. Zhen Xi

— Academic experience

2024– **Researcher**
Neurorobotics Group, University of Freiburg, Germany

Part of the project IN-CODE funded by the German Research Foundation (DFG) for developing optimization theory, methods, and software for scientific machine learning.

2022–2023 **Research intern**
Optophysiology Lab, University of Freiburg, Germany

Led a research project on the mathematical modeling of rodent decision making behavior via (inverse) reinforcement learning.

2022 **Research intern**
Straw Lab, University of Freiburg, Germany

Led a research project on developing Kalman filter auto-tuning algorithm for animal tracking, and designing event-camera-based lock-on tracker prototype, steering multiple cameras for tracking bees in the wild.

2017–2021 **Undergraduate researcher/research assistant**
State Key Laboratory of Elemento-Organic Chemistry, Nankai University, China

Led the National Training Program of Innovation and Entrepreneurship for Undergraduates project titled *Computational Simulation and Biological Verification for Different Species of Protoporphyrinogen IX Oxidase Amino Acid Interactions*, aiming at identifying key amino acid residues in protoporphyrinogen oxidase with computational quantum mechanics, molecular dynamics simulation, and graph theory.

— Teaching

Lecture

Materials of all my classes are made public at `https://haozhu10015.github.io/teaching`.

2026– **Head lecturer**
Disciplined Machine Learning, University of Freiburg

I (together with Prof. Joschka Boedecker) wrote the textbook for this course in 2026 and have been teaching it since then.

2024– **Co-lecturer**
Probabilistic Graphical Models, University of Freiburg

I developed the materials for this course in 2024 and have co-taught it with Prof. Joschka Boedecker every summer since then.

Seminar

2026 **Supervisor**
Causal World Models, University of Freiburg

2024 **Supervisor**
Causality and Reinforcement Learning, University of Freiburg

— Software

Below is a selection of my software projects. A complete list of projects I have led or contributed to is available at `https://haozhu10015.github.io/software`.

- `morew`: a mathematical optimization rewriting system in Lean
- `blvpy`: a Python package for disciplined bilevel programming based on CVXPY
- `dbcp`: a CVXPY extension for disciplined biconvex programming

- `cvxgenrust`: an embedded code generator for convex optimization problems in Rust
- `cvxrust`: a Rust implementation of the disciplined convex programming framework; under development
- `jemdoc-rs`: a Rust rewrite of the static site generator `jemdoc` for creating academic personal websites
- `rlfit`: a Python package for fitting reinforcement learning models to behavior data under multi-armed bandits
- `hiql`: hierarchical inverse Q-learning for behavior characterization

— Awards

2020 Innovative Scientific Research Award for College Students, Nankai University (Excellence Award)

2018 Asymchem Scholarship of Chemistry College, Nankai University

2017 Asymchem Scholarship of Chemistry College, Nankai University

— Invited talks

2026 **CVXPY Workshop**
Stanford University
Title: *Disciplined biconvex programming*

2024 **Bernstein Center Freiburg Seminar**
University of Freiburg
Title: *Inverse reinforcement learning in animal behavior characterization*

— Publications

The following publications are up to date as of November 2, 2026. A more frequently updated list of my publications is available at `https://haozhu10015.github.io/papers`.

Book

- H. Zhu and J. Boedecker. *Disciplined Machine Learning*, 2026. Working draft, available at `https://haozhu10015.github.io/dmlbook/`.

Journal and conference papers

- H. Zhu and J. Boedecker. Disciplined biconvex programming. *Journal of Optimization Theory and Applications*, 211:21, 2026. Preprint on *arXiv*, 2511.01813, 2025.

- S. M. B. Azad, J. Hoffmann, I. Nematollahi, H. Zhu, A. Valada, and J. Boedecker. Spectral alignment in forward-backward representations via temporal abstraction. In *Advances in Neural Information Processing Systems*, volume 40, 2026. Preprint on *arXiv*, 2603.20103, 2026.
- H. Zhu, J. Hoffmann, B. Zhang, and J. Boedecker. Fitting reinforcement learning model to behavioral data under bandits. *Frontiers in Applied Mathematics and Statistics*, 12:1762084, 2026. Preprint on *arXiv*, 2511.04454, 2025.
- M. Alyahyay, J. J. Ammer, G. Kalweit, H. Zhu, A. Adzemovic, M. Kalweit, G. Karvat, A. Schneider, Y. Tong, M. D. Döbrössy, Z. Jäckel, C. Akmese, A. Vlachos, J. Boedecker, and I. Diester. Mechanisms of premotor-motor cortex interactions during movement initiation. *Cell Reports*, 45(6):117542, 2026. Preprint on *bioRxiv*, 524944, 2023.
- B. Zhang, Y. Zhang, H. Zhu, S. Yan, T. Brox, and J. Boedecker. Constrained reinforcement learning with smoothed log barrier function. *Transactions on Machine Learning Research*, 2025.
- H. Zhu, B. De La Crompe, G. Kalweit, A. Schneider, M. Kalweit, I. Diester, and J. Boedecker. Multi-intention inverse Q-learning for interpretable behavior representation. *Transactions on Machine Learning Research*, 2024. Preprint on *arXiv*, 2311.13870, 2023.
- B. Wang, Z. Zhang, H. Zhu, C. Niu, X. Wen, and Z. Xi. The hydrogen bonding network involved Arg59 in human protoporphyrinogen IX oxidase is essential for enzyme activity. *Biochemical and Biophysical Research Communications*, 557:20–25, 2021.

Manuscripts and under review preprints

- H. Zhu and J. Boedecker. Disciplined bilevel programming. *arXiv*, 2609.00644, 2026.
- H. Zhu and J. Boedecker. Generation of custom solvers in Rust for convex optimization. *arXiv*, 2609.13875, 2026.
- W. Sheng, H. Zhu, and J. Boedecker. Probabilistic recurrent intention switching model. *arXiv*, 2605.26998, 2026.
- H. Zhu, S. Yan, J. Hoffmann, and J. Boedecker. Multi-convex programming for discrete latent factor models prototyping. *arXiv*, 2504.01431, 2025.
- K. Kleespies, P. C. Paulus, H. Zhu, F. Pargent, M. Jakob, J. Werle, M. Czisch, J. Boedecker, S. Gais, and M. Schönauer. Sleep resolves competition between explicit and implicit memory systems. *bioRxiv*, 639581, 2025.
- H. Zhu and J. Boedecker. Solving inverse problem for multi-armed bandits via convex optimization. *arXiv*, 2501.18945, 2025.
- H. Zhu, Y. Zhang, and J. Boedecker. Inverse reinforcement learning via convex optimization. *arXiv*, 2501.15957, 2025.

— References

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- *Prof. Ilka Diester*
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